toys10_06-paper-rocket

Paper Rocket Air Launcher: manufacturing manual

> **Status: design only; nothing built or pressure-tested. This is a pressurised projectile toy: do not sell before a design review against the toy standard and a lab test. The 0.3 bar limit below is a design limit, not a safety certification.**

1. What is made

A PVC pressure chamber and 19 mm barrel with a bicycle-pump connector, a spring relief valve set at about **0.29 bar gauge** (launch energy 0.5 J, range about 16 m at 42 degrees: estimates), a trigger valve and rolled-paper rockets with foam noses (templates to be printed).

No drawings exist yet; this is a plumbing assembly. Calculation: `toys10/docs/results_rocket.md` (a first draft with a 1 bar valve gave 2.8 J, five times the design limit, so the low-pressure valve is mandatory).

2. Parts (about Rs 310 at volume, estimate)

Item	Rs
PVC chamber + barrel + fittings	90
Bike-pump connector/valve	45
Relief valve (spring)	40
Trigger valve	35
Paper rocket pack + foam noses	25
Box + manual	40
Assembly + leak test	35

3. Assembly and pressure test (workshop)

1. Cut PVC chamber and barrel; deburr; solvent-weld fittings with the pipe-cement manufacturer's cure time (24 h).
2. Fit the spring relief valve (0.29 bar) in the chamber wall, trigger valve, bicycle-pump connector with a check valve.
3. **Proof test every unit** to 1.5 x relief pressure (about 0.45 bar) with water-immersion or soapy-water leak check behind a shield; no leaks, no deformation.
4. **Relief valve check:** pump until the valve opens: pressure must never exceed 0.35 bar on a calibrated gauge (reject otherwise).
5. Fire a test rocket at a safe angle with the barrel pointed away from people; measure range and compare with the estimate.
6. Pack with 6 paper rockets, foam noses, a printed instruction sheet and the warnings below.

4. Required warnings

NEVER aim at people or animals; eye protection recommended; adult supervision; 8+; no modification of the relief valve.

5. Compliance

Toys QCO / IS 9873 applicability and testing; age grade and small-parts warnings; Legal Metrology labelling; paint/ink heavy-metal report; no licensed names or logos.

Projectile safety is the whole risk: get a lab review (toy standard, pressure-vessel margin) before selling.

6. Known limits

`docs/results_rocket.md`: with a relief valve at about 0.29 bar the launch energy is 0.5 J, range about 16 m at 42° (estimate). **The first draft with a 1 bar valve gave 2.8 J, five times my placeholder energy limit; the design was changed to the low-pressure valve.** Nothing measured: the loss factor and drag are guesses. The legal energy limit must be taken from the toy standard (not verified).

Projectile safety is the whole risk: eye injuries, misuse. Needs a design review against the toy standard, a pressure-vessel safety margin, and a clear 'never aim at people' label. Consider selling only after a lab review.